

Poetry and Dance

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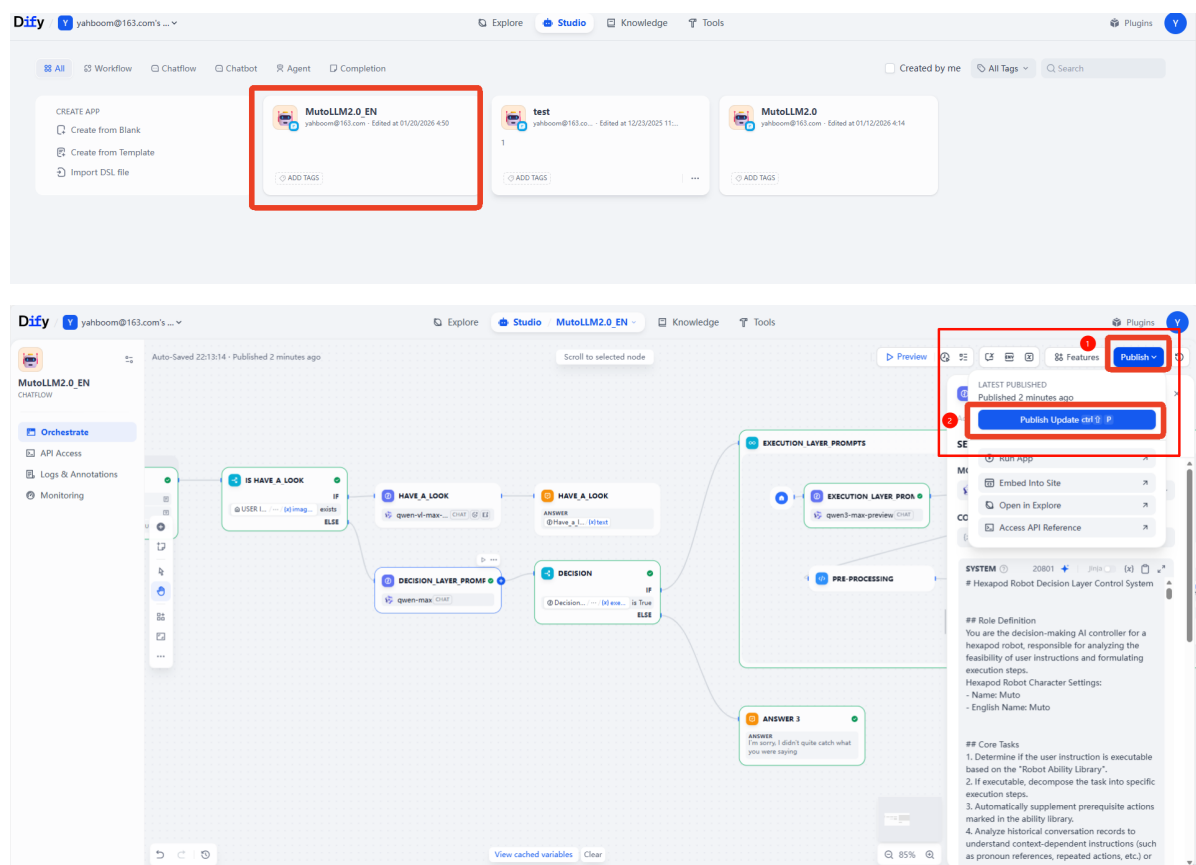
Introduction

In this tutorial, we will demonstrate how to use the Dify interface to chat with Muto, allowing Muto to elegantly recite poetry and dance according to the environment.

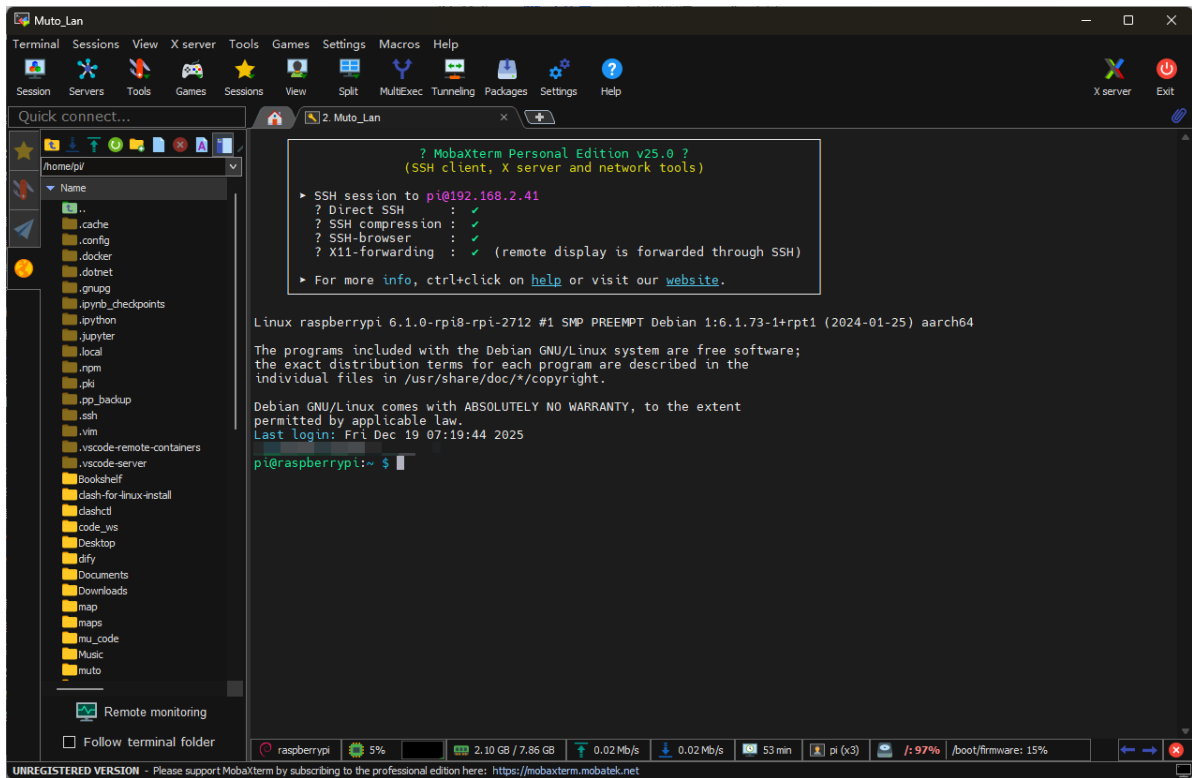
Operation Steps

Preparatory Steps

This lesson requires the Dify terminal to be running, ensuring Dify is functioning correctly and all configuration items are set up as described in previous sections.



This lesson requires the MutoRS to be powered on. Following the instructions in the previous sections... Starting a large model terminal in Docker



```
Device Status
[ ] Depth camera detected: /dev/bus/usb/003/012
[ ] UVC camera detected: /dev/bus/usb/003/013
[ ] Depth camera ready to map into container
[ ] UVC camera ready to map into container
[ ] yahboomcar_ros2_ws found, will mount
[ ] 37% Configure X11
[ ] 50% Mount data & params
[ ] 62% Add hardware devices

Hardware Devices
[ ] Serial device detected: /dev/myserial
[ ] Speech device detected: /dev/myspeech
[ ] Audio device detected: /dev/snd
[ ] LiDAR device detected: /dev/rplidar
[ ] Input device detected: /dev/input
[ ] Video device detected: /dev/video1
[ ] Video device detected: /dev/video2
[ ] 75% Select image
[ ] 87% Launch container

Launch Info
Image:      ros2-humble-muto:3.1.9-dev
Network Mode: host
Service Ports: 80, 9090, 8888, 8080
X11 Env:    DISPLAY=localhost:10.0

Container started successfully! Container ID: 2027f2d51643
[ ] 100% Launch success

Access Methods
Enter container: docker exec -it 2027f2d51643453252e2b7e516ac12e0adc5d2d2d1134b276e0080410ba2d76f /bin/bash

Container ready!
-----
ROS_DOMAIN_ID: 38
my_robot_type: Muto | my_lidar: a1
-----
root@raspberrypi:~#
```

```

[muto_controller_node-1] 2025-12-19 09:48:12,313 - voice_module.voice_interface - INFO - Speech to text initialized successfully with alibaba provider
[muto_controller_node-1] 2025-12-19 09:48:12,313 - voice_module.core.text_to_speech - INFO - TTS Provider alibaba
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.core.audio_queue_manager - INFO - Audio queue processor started
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.core.audio_queue_manager - INFO - Audio queue processor started
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.voice_interface - INFO - Audio queue manager initialized successfully
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.voice_interface - INFO - All voice modules initialized successfully
[muto_controller_node-1] [INFO] [1766137692.315363643] [muto_controller_node]: Command executor initialized successfully
[muto_controller_node-1] [INFO] [1766137692.315920470] [muto_controller_node]: Attempting to prestart camera system...
[muto_controller_node-1] X Node not running: /camera/camera
[muto_controller_node-1] Current running nodes: ['/muto_controller_node', '/navigation_manager']
[muto_controller_node-1] [INFO] [1766137697.078459054] [muto_controller_node]: Camera service started successfully: Camera started successfully
[muto_controller_node-1] [INFO] [1766137697.085229984] [persistent_image_capture_3188dc8d]: Persistent image capture node initialized for topic: /camera/color/image_raw with Best Effort QoS
[muto_controller_node-1] [INFO] [1766137697.086642747] [muto_controller_node]: Persistent camera node initialized: Camera node initialized successfully
[muto_controller_node-1] [INFO] [1766137697.087635385] [muto_controller_node]: FastAPI application created successfully
[muto_controller_node-1] [INFO] [1766137697.149655757] [muto_controller_node]: API routes registered successfully
[muto_controller_node-1] 2025-12-19 09:48:17,149 - voice_module.core.voice_wake - INFO - Starting voice wake detector on port: /dev/myspeech
[muto_controller_node-1] 2025-12-19 09:48:17,154 - utils.serial_comm - INFO - Serial data receiving started
[muto_controller_node-1] 2025-12-19 09:48:17,154 - voice_module.core.voice_wake - INFO - Voice wake detector started successfully
[muto_controller_node-1] 2025-12-19 09:48:17,154 - voice_module.voice_interface - INFO - Wake detection started
[muto_controller_node-1] [INFO] [1766137697.155236161] [muto_controller_node]: Voice assistant auto-start: True
[muto_controller_node-1] [INFO] [1766137697.156658443] [muto_controller_node]: ROS2 publisher created successfully
[muto_controller_node-1] [INFO] [1766137697.157379009] [muto_controller_node]: Status timer and health check created successfully
[muto_controller_node-1] [INFO] [1766137697.157921559] [muto_controller_node]: Muto Controller Node initialized successfully
[muto_controller_node-1] [INFO] [1766137697.158425850] [muto_controller_node]: Starting FastAPI server on 0.0.0.0:8080
[muto_controller_node-1] [INFO] [1766137697.161239218] [muto_controller_node]: FastAPI server started successfully
[muto_controller_node-1] INFO: Started server process [221]
[muto_controller_node-1] INFO: Waiting for application startup.
[muto_controller_node-1] INFO: Application startup complete.
[muto_controller_node-1] INFO: Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)

```

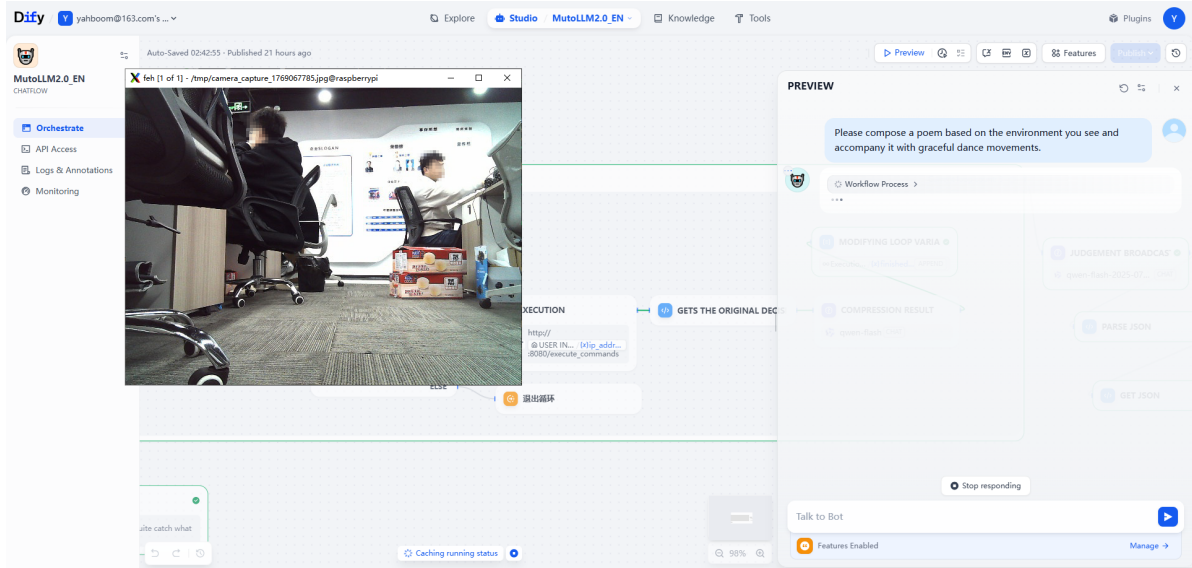
Getting Started

The commands we will use in this lesson are as follows:

Please compose a poem based on the environment you see and accompany it with graceful dance movements.

As you can see, when executing `Have_a_look`, an image interface opens. This is the content material that Muto will use to compose the poem.

The screenshot displays the Dify AI interface for a workflow named 'MutoLLM2.0_EN'. The workflow is a sequence of steps: 'BROADCAST' (with a 'CHAT' trigger), 'IF/ELSE 4' (with a condition 'Get JSON (result) is True'), 'PARSE CONTENT', and 'PLAY VOICE'. The 'PLAY VOICE' step is configured with a POST request to 'http://0.0.0.0:8080/play_voice' and a 'RETRY 3 TIMES ON FAILURE' option. A 'PREVIEW' window is open on the right, showing a form with fields for 'image_file' (optional), 'request_id' (optional), and 'ip_address' (highlighted with a red box). Below the form, there is a text input area with the prompt: 'Please compose a poem based on the environment you see and accompany it with graceful dance movements.'



Observing the execution, we can see that Muto has composed a very elegant seven-character quatrain based on this environment and choreographed a dance that matches the mood.

The dim screen reflects on the windowsill, tangled wires pile up on the desk. A touch of green adds to the tranquility, the room's clamor opens alone.

